

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks: 30

Annexure A

Dry Run Evaluation

Code of Conduct:

*I M. Arshini 192424322 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: M. Arshini

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1. Question 18 - Minimax with Deeper Tree

Consider a MAX root with two MIN children. Each MIN child has two MAX children. The leaf values from left to right are: 3,5 | 2,9 | 4,6 | 1,7.

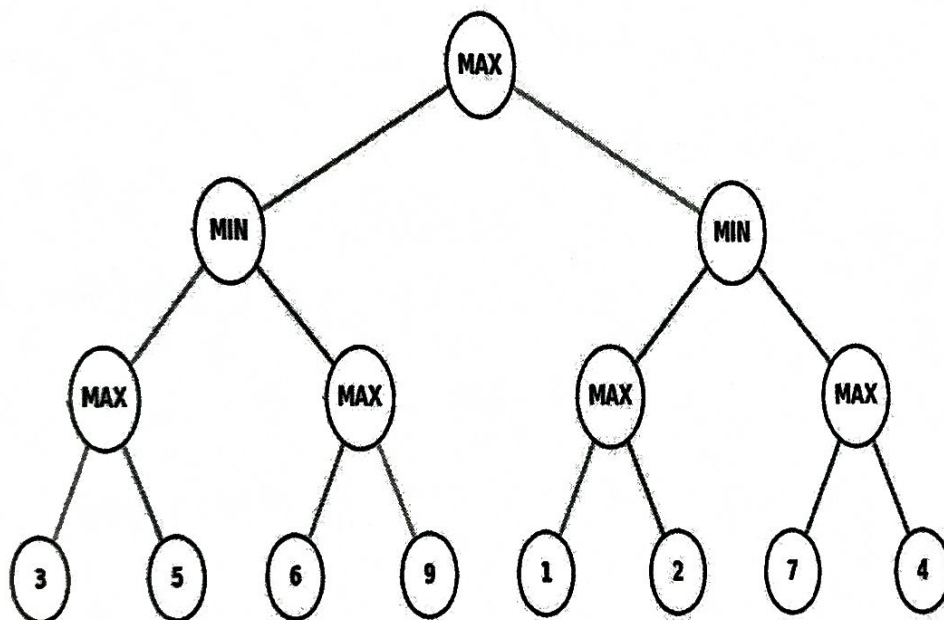
Perform a complete Minimax dry run and identify the optimal move.

Node	Children Values	Value
1		
2		
3		
4		
5		
6		
7		

2. Question 19 - Alpha-Beta Pruning

Evaluate the displayed game tree from left to right using Alpha-Beta pruning.

Show alpha and beta values, identify pruned nodes and determine the final minimax value.



Node	Alpha	Beta	Pruned
1			
2			
3			
4			
5			

3. Question 20 - Evaluation Function

An AI game-playing agent uses $E(s)=2M+3K+5Q+P$.

For S1: $M=4, K=2, Q=1, P=5$. For S2: $M=3, K=5, Q=2, P=4$.

Calculate both evaluation scores and determine which state MAX should select.

Minimax with Deeper tree

Given

max roots has two min children

Each min node has two max children

Leaf values - from left to right

3, 5, 2, 9, 4, 6, 1, 7

Step 1:- Evaluate max nodes

$$\text{MAX}(3, 5) = 5$$

$$\text{MAX}(2, 9) = 9$$

$$\text{MAX}(4, 6) = 6$$

$$\text{MAX}(1, 7) = 7$$

Step 2:- Evaluate MIN nodes :-

Left min:

$$\text{MIN}(5, 9) = 5$$

Right- MIN:-

$$\text{MIN}(6, 7) = 6$$

step 2:- evaluate root max

$$\text{MAX}(5, 6) = 6$$

Node	children values	values
MAX 1	3, 5	5
MAX 2	2, 9	9
MIN 1	5, 9	5
MAX 3	4, 6	6
MAX 4	1, 7	7
MIN 2	6, 7	6
Rootmax	5, 6	6

optimal move = right subtree.

~~Alpha Beta pruning~~

The tree is evaluated from left to right

Initial Values

$$\alpha = -\infty, \quad \beta = +\infty$$

Left MIN subtree

First max node:

$$\text{MAX}(3, 5) = 5$$

MIN:

$B = 5$

Now, Evaluate the second max node with $B = 5$

first value: 6

Since:

$$N = 6 \geq B = 5$$

the remaining node is pruned

So:

$$\max(6, 9) = 6$$

Left MIN:

$$\min(5, 6) = 5$$

Node gets: $d = 5$

Right MIN subtree:

first max node:

$$\max(1, 2) = 2$$

At the right MIN node:

$$B = 2$$

But root already has:

$$d = 5$$

Since: $B = 2 \leq d = 5$

The entire remaining max subtree (min) is pruned

$$\min(2, \max(7, 8)) = 2$$

Root max

$$\text{MAX}(5, 2) = 5$$

Alpha - Beta

Node	α	β	Action/pruning
Root max	$-\infty \rightarrow 5$	$+\infty$	left subtree evaluated
Left MIN	$-\infty$	$+\infty \rightarrow 5$	continue
MAX (3, 5)	$-\infty \rightarrow 5$	5	value = 5
MAX (6, 9)	$5 \rightarrow 6$	5	9 pruned
Right MIN	5	$+\infty \rightarrow 2$	(7, 4) pruned

2) Evaluation function:-

Given

$$E(S) = 2M + 3K + 5Q + P$$

state S1

$$M=4, K=2, Q=1, P=5$$

substitute

$$E(S1) = 2(4) + 3(2) + 5(1) + 5$$

$$= 8 + 6 + 5 + 5$$

$$E(S1) = 24$$

State S2

given

$$M=3, K=5, Q=2, P=4$$

$$E(S2) = 2(3) + 3(5) + 5(2) + 4$$

$$= 6 + 15 + 10 + 4$$

$$E(S_2) = 35$$

comparison

$$E(S_1) = 24$$

$$E(S_2) = 35$$

Since MAX choose, the state with the higher evaluation score:

$$35 > 24$$

max should select state S_2



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